A. Let $h(u,v) : U' \to V' = (u\cos v, u\sin v)$ where $U' = U$ and $V' = V \setminus \{(s,t) : t = 0, s = -1\}$.

B. We first compute g_x. We compute that $x_u = (\cos v, \sin v, 1)$ and $x_v = (-u\sin v, u\cos v, 0)$. Then, $x_u \cdot x_u = \cos^v + \sin^2 v + 1 = 2$, $x_u \cdot x_v = -u\cos v \sin v + u\cos v \sin v = 0$ and $x_v \cdot x_v = u^2\cos^v + u^2\sin^2 v = u^2$. So, $g_x = \begin{bmatrix} 2 & 0 \\ 0 & u^2 \end{bmatrix}$. We now compute g_y. We compute that

$$g_y = \begin{bmatrix} 1 + \frac{s^2}{s^2+t^2} & \frac{st}{s^2+t^2} \\ \frac{st}{s^2+t^2} & 1 + \frac{t^2}{s^2+t^2} \end{bmatrix}$$

Evaluating at h, g_y becomes $g_y = \begin{bmatrix} 1+\cos^2 v & \cos v \sin v \\ \cos v \sin v & 1 + \sin^2 u \end{bmatrix}$. We then compute $\frac{\partial h}{\partial \boldsymbol{u}} = \begin{bmatrix} \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{bmatrix} = \begin{bmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{bmatrix}$. Next, we compute that

$$\frac{\partial h}{\partial \boldsymbol{u}}^T g_y = \begin{bmatrix} \cos v + \cos^3 v + \sin^3 v \cos v & \cos^2 v \sin v + \sin v + \sin^3 v \\ -u\sin v - u\cos^2 v \sin v + u\cos^2 v \sin v & u\cos v \end{bmatrix}$$

We multiply this result by $\frac{\partial h}{\partial \boldsymbol{u}}$:

$$\begin{bmatrix} \cos v + \cos^3 v + \sin^3 v \cos v & \cos^2 v \sin v + \sin v + \sin^3 v \\ -u\sin v - u\cos^2 v \sin v + u\cos^2 v \sin v & u\cos v \end{bmatrix} \begin{bmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{bmatrix}$$

and find that

$$\begin{bmatrix} \cos^2 v + \cos^4 v + 2\sin^2 v \cos^2 v + \sin^2 v + \sin^4 v & 0 \\ 0 & u^2\sin^2 v + u^2\cos^2 v \end{bmatrix}$$

This simplifies to

$$\begin{bmatrix} (\cos^2 v + \sin^2 v)^2 + 1 & 0 \\ 0 & u^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & u^2 \end{bmatrix}$$

C. Under x, $x_1 = u\cos v$. This is greater than or equal to 0 when $u \geq 0$ and $\cos v \geq 0$. $\cos v \geq 0$ when $v \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. So, the corresponding Q of U is $(0,1) \times (-\frac{\pi}{2}, \frac{\pi}{2})$. The area of R as computed using x and Q is

$$\iint_R 1\, dA = \iint_Q \sqrt{\det g_x}\, du\, dv = \int_{-\pi/2}^{\pi/2} \int_0^1 \sqrt{\det g_x}\, du\, dv = \sqrt{2} \int_{-\pi/2}^{\pi/2} \left(\frac{u^2}{2}\Big|_0^1\right) dv = \frac{\sqrt{2}\pi}{2}.$$

Under y, $x_1 = s$, so we want $s \geq 0$. This corresponds to the region $P = \{(s,t) : 0 < s^2 + t^2 < 1, s \geq 0\}$. This is the right half of the unit circle. We can integrate using polar coordinates.

$$\iint_R 1\, dA = \iint_P \sqrt{\det g_y}\, ds\, dt = \sqrt{2} \iint_P 1\, ds\, dt = \int_{-\pi/2}^{\pi/2} \int_0^1 r\, dr\, d\theta = \frac{\sqrt{2}\pi}{2}.$$